understood that space is very much like matter – space, as he had said himself, has its own nature, ‘a certain physics’ – I suspect he would not have hesitated to deduce that space, too, can only be made of elementary chunks that cannot be divided. We are perhaps just following in the footsteps of Democritus.

I certainly don’t mean to imply that the physics of two millennia was useless, that experiments and mathematics are pointless and that Democritus could be as convincing as modern science. Obviously not. Without experiments and mathematics, we would never have understood what we have understood. Yet we develop our conceptual schema for understanding the world by exploring new ideas but also by building on the powerful intuitions of giant figures from the past. Democritus is one of them, and we discover the new sitting on his titanic shoulders.

But let us return to quantum gravity.

Spin networks

The graphs which describe the quantum states of space are characterized by a volume v for every node and a half-integer j for every line. A graph with this additional information is called a spin network (figure 6.4). (Half-integers in physics are called ‘spin’ because they appear in the quantum mechanics of spinning objects.) A spin network represents a quantum state of the gravitational field: a quantum state of space; a granular space in which area and volume are discrete. Fine-mesh grids are used elsewhere in physics to approximate continuous space. Here, there is no space continuum to approximate: space is genuinely granular.

The crucial difference between photons (the quanta of the electromagnetic field) and the nodes of the graph (the quanta of gravity) is that photons exist in space, whereas the quanta of gravity constitute space themselves. Photons are characterized by ‘where they are’.[fn37] Quanta of space have no place to be in, because they are themselves that place. They have only one piece of information which characterizes them spatially: information about which other quanta of space they are adjacent to, which one is next to which other. This information is expressed by the links in the graph. Two nodes connected by a link are two nodes in proximity. They are two grains of space in contact with each other: this ‘touching’ constructs the structure of space.